A Digital Twin Framework for Data Centre Cooling Using Industrial Waste Heat and Water-Vapour-Assisted Cooling

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Abstract. Data centers use great quantities of electricity and their cooling systems form a significant portion of their overall energy demand. Meanwhile, waste heat from industrial processes can be recovered for useful applications. This paper proposes a digital twin framework integrating real-time data center monitoring, industrial waste heat recovery and water-vapour-assisted cooling. The digital twin combines sensor data, system modeling and prediction analysis to allow cooling optimization under varying workloads and environmental conditions. The industrial heat recovered can be used to support suitable thermal cooling technologies, e.g. absorption cooling or desiccant based dehumidification, depending on the temperature available and system configuration. It introduces a conceptual architecture and an illustrative evaluation framework. The proposed approach aims to reduce the demand for conventional cooling energy, while increasing operational visibility and energy utilization. Quantitative performance improvements should be validated by simulation or experimental measurements.

Keywords: Digital Twin · Data Centre Cooling · Waste Heat Recovery · Water Vapour · Energy Efficiency

1 Introduction

The rapid growth of cloud computing, artificial intelligence and high performance computing has increased the demand for data center infrastructure. These facilities need dependable cooling to keep the servers at a temperature within acceptable operating limits. Traditional cooling systems consume a lot of energy, which results in higher operating costs and a larger environmental footprint.

Simultaneously, industrial plants frequently reject thermal energy that could potentially be recovered instead of rejected to the environment. The integration of such waste heat with suitable cooling technologies gives a chance to improve overall energy utilization.

A digital twin is a virtual representation of a physical system that makes use of operational data to enable monitoring, simulation, prediction, and decision-making. For data centers, a digital twin can combine data on server loads, temperatures, humidity, cooling equipment and energy use.

In this paper, a conceptual digital twin framework linking data center cooling with industrial waste heat recovery and water-vapour-assisted cooling is presented. Our main contributions are:

- A conceptual architecture connecting sensors, a digital twin, waste heat recovery, and cooling equipment.
- A modelling approach for monitoring and predicting cooling requirements.
- An illustrative evaluation framework for comparing cooling energy demand under different operating conditions.

2 Background and Related Concepts

2.1 Digital Twins in Data Centres

Digital twins use data from physical equipment to maintain a virtual model of system behaviour. In a data centre, the model can represent server loads, airflow, temperature distribution, and cooling-system performance. Predictions from the model can support operational planning and control.

2.2 Industrial Waste Heat Recovery

Industrial processes release heat at different temperature levels. The suitability of this heat for cooling applications depends on its temperature, availability, variability, and the technology selected. Absorption cooling may use thermal energy to produce a cooling effect, while desiccant systems can use heat to regenerate moisture-removal materials.

2.3 Water-Vapour-Assisted Cooling

Evaporative and desiccant-assisted systems use the interaction between water vapour, heat, and air to support cooling or humidity control. Their effectiveness depends on ambient humidity, water availability, airflow, and equipment design. These technologies must be selected carefully to maintain suitable data centre operating conditions.

3 Proposed System Architecture

The proposed framework consists of a physical data center, a sensor and building management system (BMS), a digital twin, an optimization module, an industrial waste heat interface and a cooling subsystem.

Sensors monitor server inlet air temperatures, coolant temperatures, relative humidity, air flow, electrical power, and waste heat. The collected measurements are transmitted to the digital twin, which estimates the current system state and predicts the cooling requirements.

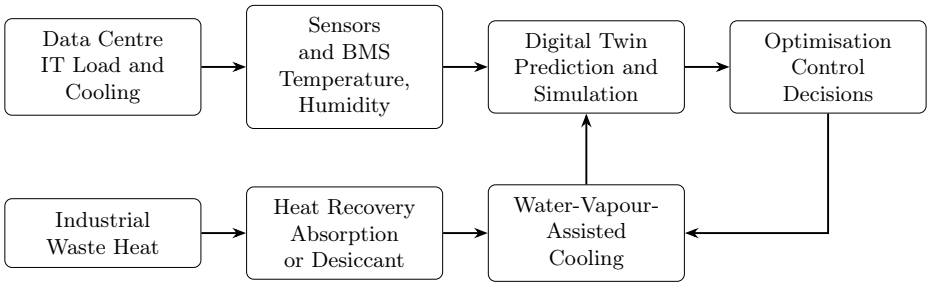


Fig. 1. Conceptual digital twin architecture integrating industrial waste heat recovery and water-vapour-assisted cooling.

The optimization module uses these estimates to help control decisions. Available industrial heat may be utilized for a compatible heat-driven cooling or desiccant regeneration process depending upon equipment limitations and operating conditions. The cooling is then controlled using water-vapour-assisted cooling depending on the cooling requirement and the ambient conditions.

4 Mathematical Model

4.1 Cooling Energy Demand

Let the total electrical energy consumption of a data centre be represented by

$$E_{\text{total}} = E_{\text{IT}} + E_{\text{cooling}} + E_{\text{other}}, \tag{1}$$

where E_{IT} is the energy used by computing equipment, E_{cooling} is the electrical energy used by cooling equipment, and E_{other} represents other facility loads.

The cooling energy reduction can be estimated as

$$R_{\text{cooling}} = \frac{E_{\text{baseline}} - E_{\text{proposed}}}{E_{\text{baseline}}} \times 100\%, \tag{2}$$

where E_{baseline} and E_{proposed} are the cooling energy consumptions of the baseline and proposed systems, respectively, measured over the same period.

4.2 Recovered Heat

The thermal power recoverable from an industrial stream can be estimated using

$$\dot{Q}_{\text{recovered}} = \dot{m} c_p (T_{\text{in}} - T_{\text{out}}), \tag{3}$$

where $\dot{m}$ is the mass flow rate, c_p is the specific heat capacity, and T_{in} and T_{out} are the inlet and outlet temperatures of the heat-transfer stream.

The amount of usable heat is limited by losses in the heat exchanger, the temperature of the source and the requirements of the cooling technology chosen. Therefore, the digital twin should estimate usable heat, instead of assuming that all available industrial waste heat can be recovered.

4.3 Digital Twin State Estimation

A simplified state-space representation can be written as

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t, \mathbf{d}_t) + \mathbf{w}_t, \tag{4}$$

$$\mathbf{y}_t = h(\mathbf{x}_t) + \mathbf{v}_t, \tag{5}$$

where $\mathbf{x}_t$ represents the system state, $\mathbf{u}_t$ represents control inputs, $\mathbf{d}_t$ represents external disturbances such as IT workload and ambient conditions, and $\mathbf{y}_t$ represents sensor measurements. The terms $\mathbf{w}_t$ and $\mathbf{v}_t$ represent process and measurement noise, respectively.

The functions f and h must be defined and calibrated for the specific data centre and cooling equipment.

5 Evaluation Framework

The framework can be tested for a conventional cooling configuration and the proposed integrated approach. Both the cases should be assessed on similar IT workloads, ambient conditions and operating periods.

Relevant measurements include cooling electricity consumption, total facility energy, server inlet temperature, humidity, recovered thermal energy, water consumption, and operating cost.

5.1 Illustrative Cooling Energy Comparison

Figure 2 shows a hypothetical normalised comparison. The values are illustrative only and do not represent measured or simulated results.

5.2 Illustrative Waste Heat Availability

Figure 3 illustrates how the usable fraction of an industrial heat source might vary with source temperature. The curve is conceptual and must be replaced by measurements or a validated thermodynamic model.

5.3 Evaluation Metrics

The following metrics can be used to assess the proposed framework.

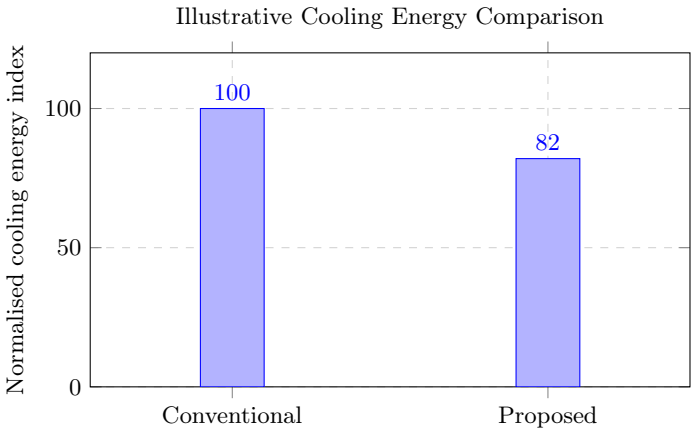


Fig. 2. Hypothetical normalised cooling energy comparison. The illustrative index of 82 for the proposed case is not a validated performance result.

Table 1. Evaluation metrics for the proposed framework.

Metric	Purpose
Cooling energy	Measures electricity consumed by cooling equipment.
Total facility energy	Measures the overall energy demand of the data centre.
Server inlet temperature	Checks whether equipment remains within its operating limits.
Humidity	Evaluates moisture control and potential condensation risks.
Recovered heat	Quantifies thermal energy transferred from the industrial source.
Water consumption	Measures water required by evaporative or vapour-assisted processes.
Control response time	Measures how quickly the system responds to workload or environmental changes.

6 Discussion

The proposed framework addresses data-driven monitoring and the potential reuse of industrial thermal energy. A digital twin can help to estimate cooling demand, identify changing operating conditions and evaluate alternative control strategies before they are applied to physical equipment.

However, the waste heat may not be useful for all cooling processes. The result depends on the temperature and stability of the heat source, the performance coefficient of the cooling equipment , heat exchanger losses and the cooling demand. Water use, humidity control and risk of condensation also need to be considered.

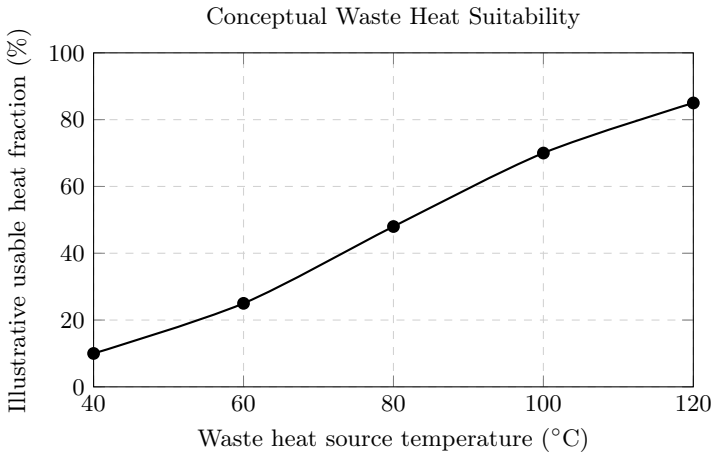


Fig. 3. Conceptual relationship between source temperature and usable heat fraction; illustrative values only.

Initially, the framework should be in a monitoring or advisory mode. Implementation of automated control should only be considered after model validation and definition of appropriate safety constraints.

7 Limitations and Future Work

The paper is a conceptual framework, not a validated prototype. The graphs shown are for illustration only and are not an actual energy saving. The proposed system needs to be implemented and evaluated either with real sensor data, calibrated thermal model or a suitable simulation environment.

Future work should develop a detailed digital twin, integrate measured industrial heat profiles, and compare suitable cooling technologies for different workloads and weather conditions. Experiments should measure energy consumption, water consumption, temperature stability and overall system efficiency. A life-cycle assessment could further be explored for environmental benefits and trade-offs.

8 Conclusion

We proposed a digital twin framework for data center cooling in this paper, which integrates physical-system monitoring, predictive modeling, industrial waste heat recovery and water-vapour-assisted cooling. The framework provides a conceptual basis for coordination of cooling demand with available thermal energy while tracking temperature, humidity, and energy consumption.

The potential benefits must be validated by simulation or experimental testing. In future implementation, energy savings and water requirements need to be quantified, while ensuring reliable and safe data center operation.

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